

Tutorial on

Programming Novel AI Accelerators for Scientific Computing

ISC26
June 22, 2026

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Matthew Povey (SambaNova), Moritz Thüning (Tenstorrent)

Tutorial Agenda

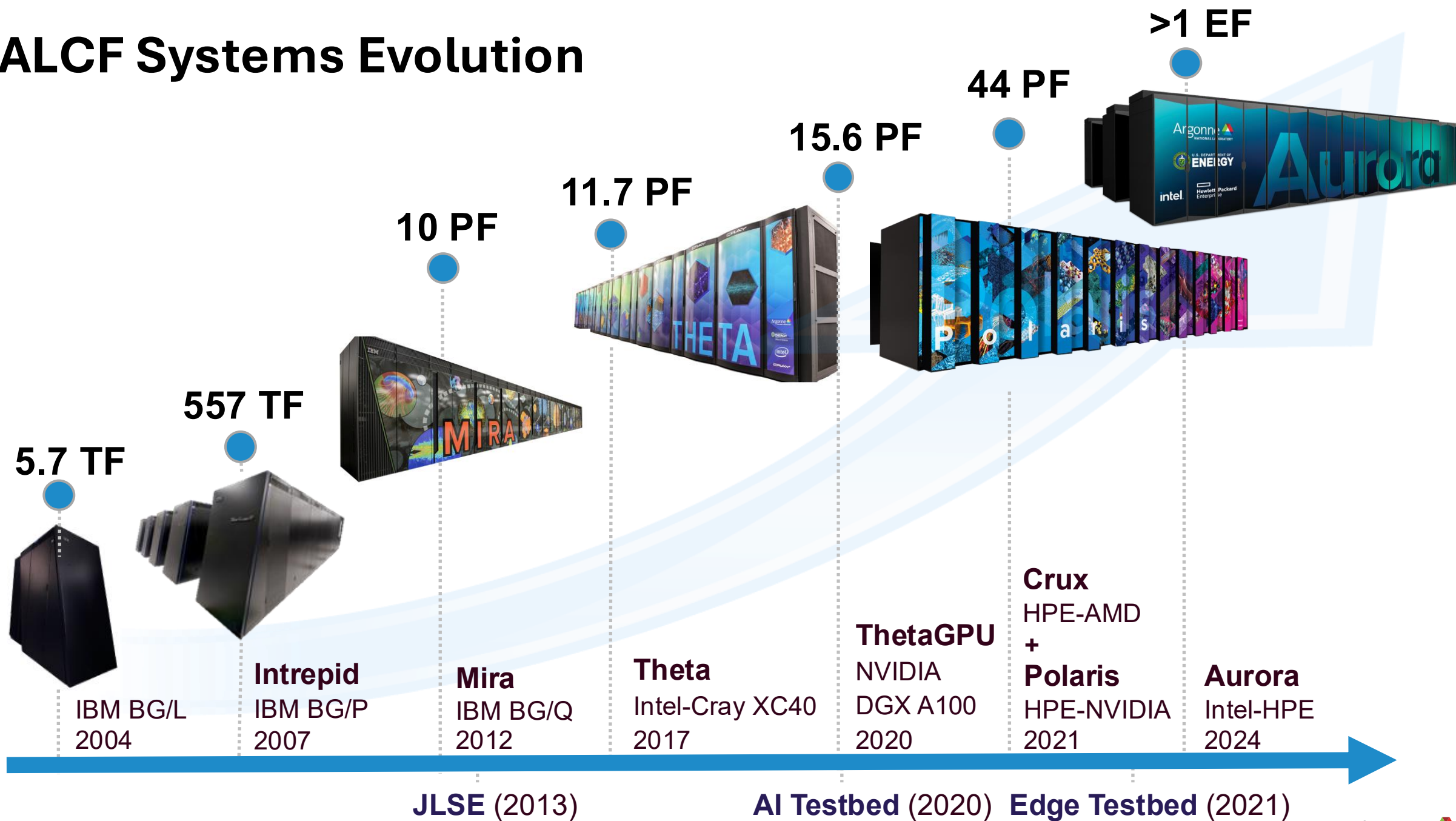
<https://github.com/argonne-lcf/ALCF-AIAccelerators-tutorial/tree/ISC26>

<https://tinyurl.com/anl-isc26>

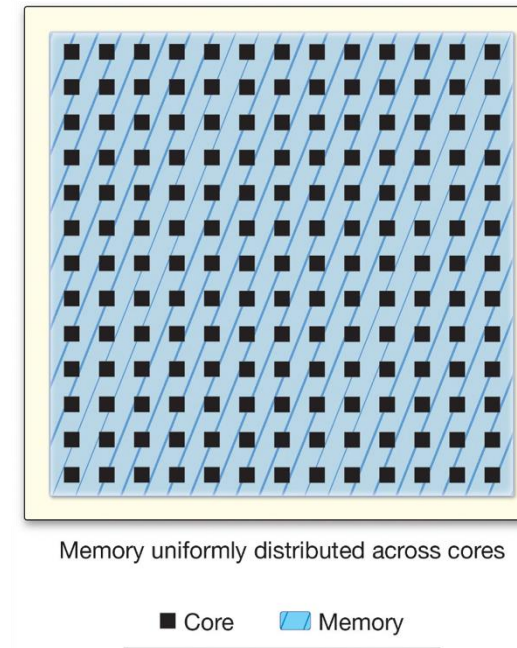
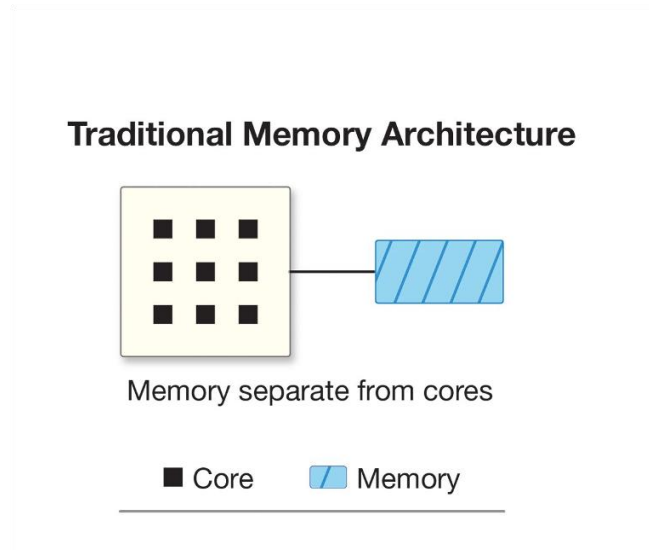
Agenda

Time	Topic	Duration (minutes)	Speaker
2:00 PM - 2:15 AM	Welcome and Overview of the ALCF AI Testbed	15	Murali (ANL)
2:15 PM - 3:15 PM	Cerebras [AI Slides , SDK Slides]	60	Sylvia (Cerebras)
3:15 PM - 4:00 PM	Sambanova [Slides]	45	Matthew (SambaNova)
04:00 PM - 04:30 PM	Coffee Break	30	
04:30 PM - 05:15 PM	Tenstorrent [Slides]	45	Moritz (Tenstorrent)
05:15 PM - 06:00 PM	Hands-on and Q&A	45	Varuni, Murali (ANL)

ALCF Systems Evolution



Von Neumann vs Data flow architectures



- Limitations of Traditional Architectures
- Heavy data movement leads to Increased Energy Cost in GPUs

- Rise of domain-specific dataflow inspired architectures
- Interleaving of compute and memory units
- Routing data through the compute elements

ALCF AI Testbed

<https://www.alcf.anl.gov/alcf-ai-testbed>



Cerebras



SambaNova



Groq



Graphcore



Tenstorrent

- Infrastructure of next-generation machines with hardware accelerators customized for artificial intelligence (AI) applications.
- Provide a platform to evaluate usability and performance of machine learning based HPC applications running on these accelerators.
- The goal is to better understand how to integrate AI accelerators with ALCF's existing and upcoming supercomputers to accelerate science insights

ALCF AI Testbed

ALCF AI Testbed Systems are in production and available for allocations to the research community

Training

- Cerebras



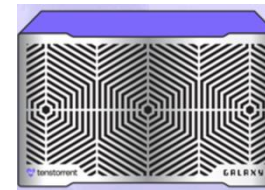
Cerebras CS-3
4 WSE ((wafer scale engine)

Inference

- SambaNova
- Groq
- Cerebras
- Tenstorrent



4 nodes of
16 SN40L RDUs



2 Tenstorrent Galaxy
32 wormhole chips



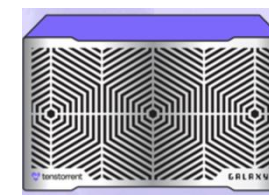
Cerebras CS-3 – 4 WSE
(upcoming support)

Low-level SDK

- Cerebras
- Tenstorrent



Cerebras CS-3



Tenstorrent Galaxy

Getting Started on ALCF AI Testbed

Available for Allocations

- Cerebras CS-3
- Sambanova Inference – Metis SN40L (Available for all ALCF users via Inference Service Endpoints)
- GroqRack
- Graphcore Bow Pod64

[AI Testbed User Guide](#)

Director's Discretionary (DD) awards

- Scaling code
- Preparing for future computing competition
- Scientific computing in support of strategic partnerships.

Allocation Request Form

<https://www.alcf.anl.gov/science/directors-discretionary-allocation-program>

NAIRR Pilot

Aims to connect U.S. researchers and educators to computational, data, and training resources needed to advance AI research and research that employs AI.

<https://nairrpilot.org/>

Machines > AI Testbed

<https://docs.alcf.anl.gov/ai-testbed/getting-started/>

ALCF AI Testbed



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How to Get Access
Getting Started
How to Contribute to Documentation

Does this doc need an update?
[Open an Issue on GitHub](#)

ALCF User Guides

Getting Started
Contribute to User Guides

User Support

Submit a Ticket
Get Help & Connect >

Machines

Aurora >
AI Testbed ✓
Cerebras >
Graphcore >
Groq >
SambaNova >
Data Management >
Crux >
Polaris >
Sophia >

Running Jobs with PBS

Example Job Scripts
Machine Reservations

Data Management

File System & Storage >
Transfer & Sharing >

Services

Inference Endpoints
JupyterHub
Continuous Integration >

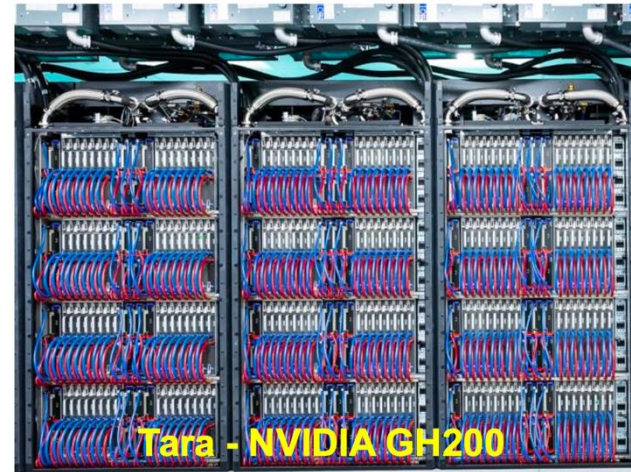
The [ALCF AI Testbed](#) houses some of the most advanced AI accelerators for scientific research.

The goal of the testbed is to enable explorations into next-generation machine learning applications and workloads, enabling the ALCF and its user community to help define the role of AI accelerators in scientific computing and how to best integrate such technologies with supercomputing resources.

The AI accelerators complement the ALCF's current and next-generation supercomputers to provide a state-of-the-art computing environment that supports pioneering research at the intersection of AI, big data, and high performance computing (HPC).

The platforms are equipped with architectural features that support AI and data-centric workloads, making them well suited for research tasks involving the growing deluge of scientific data produced by powerful tools, such as supercomputers, light sources, telescopes, particle accelerators, and sensors. In addition, the testbed will allow researchers to explore novel workflows that combine AI methods with simulation and experimental science to accelerate the pace of discovery.

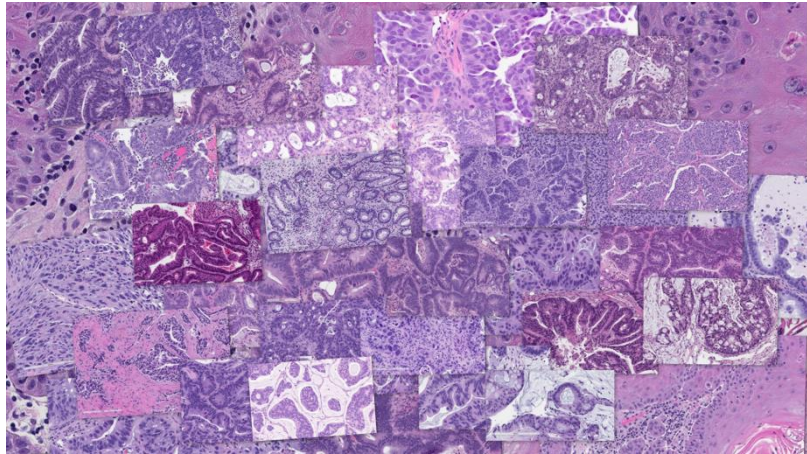
ALCF IS DEPLOYING DIVERSE INFERENCE SYSTEMS FOR SCIENCE



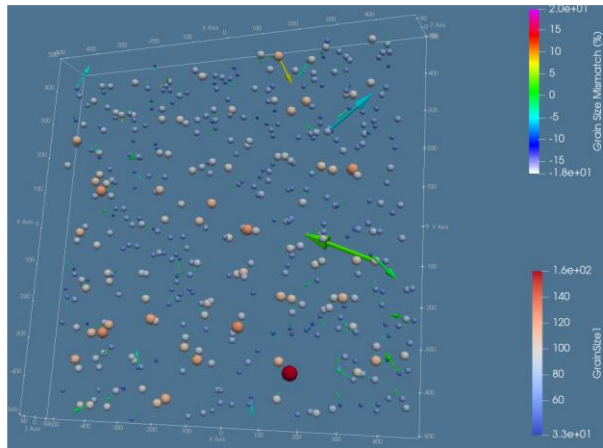
Coming soon:
**Tenstorrent Galaxy,
Minerva, Tara (Nvidia)**

<https://docs.alcf.anl.gov/services/inference-endpoints/>

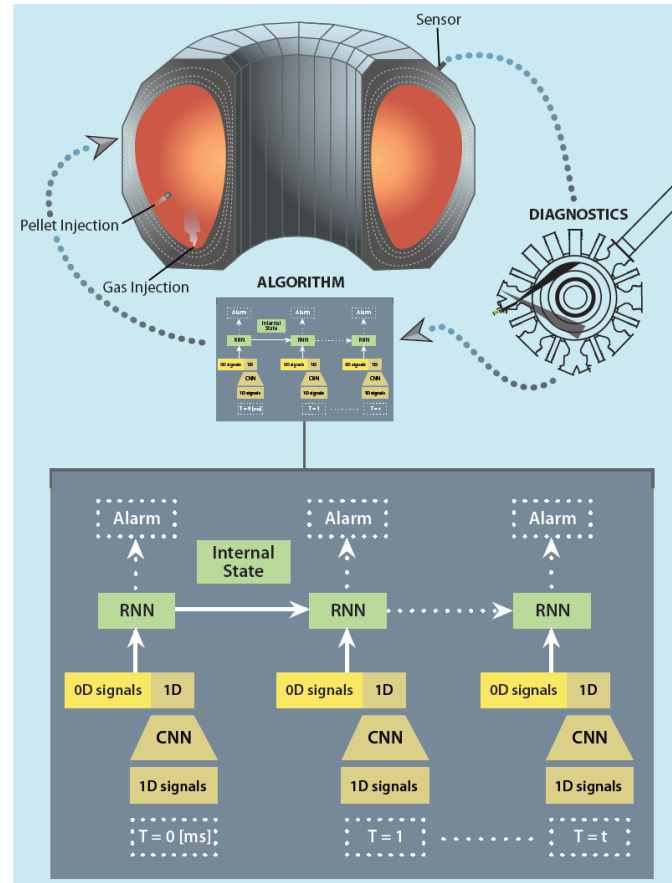
AI for Science and HPC applications on AI Testbed



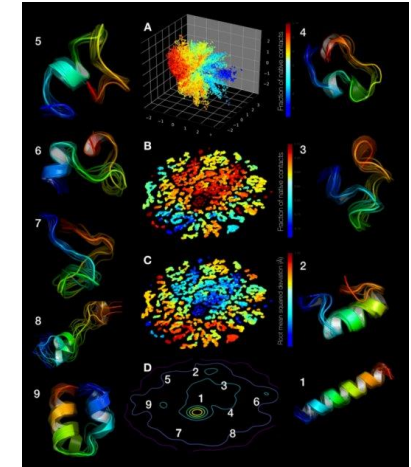
Cancer drug response prediction
(Credit: Candle)



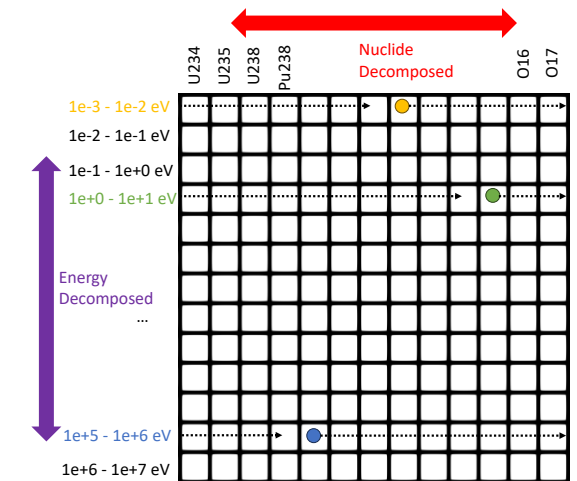
Imaging Sciences-Braggs Peak
(Credit: Z. Liu)



Tokamak Fusion Reactor operations
(Credit: K. Felker)



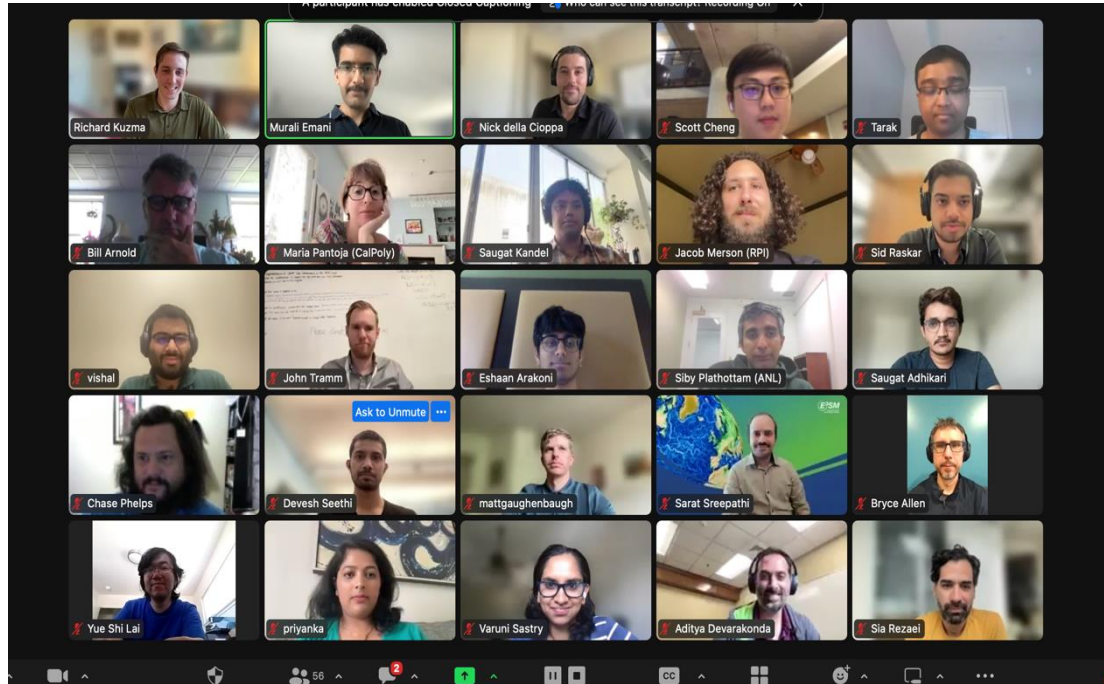
Protein-folding (Image: NCI)



Monte Carlo Particle Transport for
Reactor Simulation (Credit: J. Tramm)

and more..

AI Testbed Community Engagement



Tutorials at venues such as SC, ISC on “Programming Novel AI accelerators for Scientific Computing”

- NAIRR Annual meeting, SC25, SCA-HPC Asia

Upcoming:

- **ISC26, June 22, 2026**
- Cerebras CS-3 training (TBA)
- SC26 (Nov 26)

- AI training workshops

<https://www.alcf.anl.gov/ai-testbed-training-workshops>

- ATPESC Training
- Introduction to AI-driven Science on Supercomputers

Useful Links

ALCF AI Testbed

- Overview: <https://www.alcf.anl.gov/alcf-ai-testbed>
- Guide: <https://docs.alcf.anl.gov/ai-testbed/getting-started/>
- Training:
 - <https://www.alcf.anl.gov/ai-testbed-training-workshops>
 - <https://github.com/argonne-lcf/ALCF-AIAccelerators-tutorial/tree/ISC26>
- Allocations: <https://www.alcf.anl.gov/science/directors-discretionary-allocation-program>
- Support: support@alcf.anl.gov

Reach out for any follow up discussions:

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Thank You

- This research was funded in part and used resources of the Argonne Leadership Computing Facility (ALCF), a DOE Office of Science User Facility supported under Contract DE-AC02-06CH11357.
- Murali Emani, Michael Papka, William Arnold, Varuni Sastry, Sid Raskar, Krishna Teja-Chitty Venkata, Rajeev Thakur, Ray Powell, John Tramm, and many others have contributed to this material.
- Our current AI testbed system vendors – Cerebras, SambaNova, Tenstorrent, Graphcore, Groq. There are ongoing engagements with other vendors.